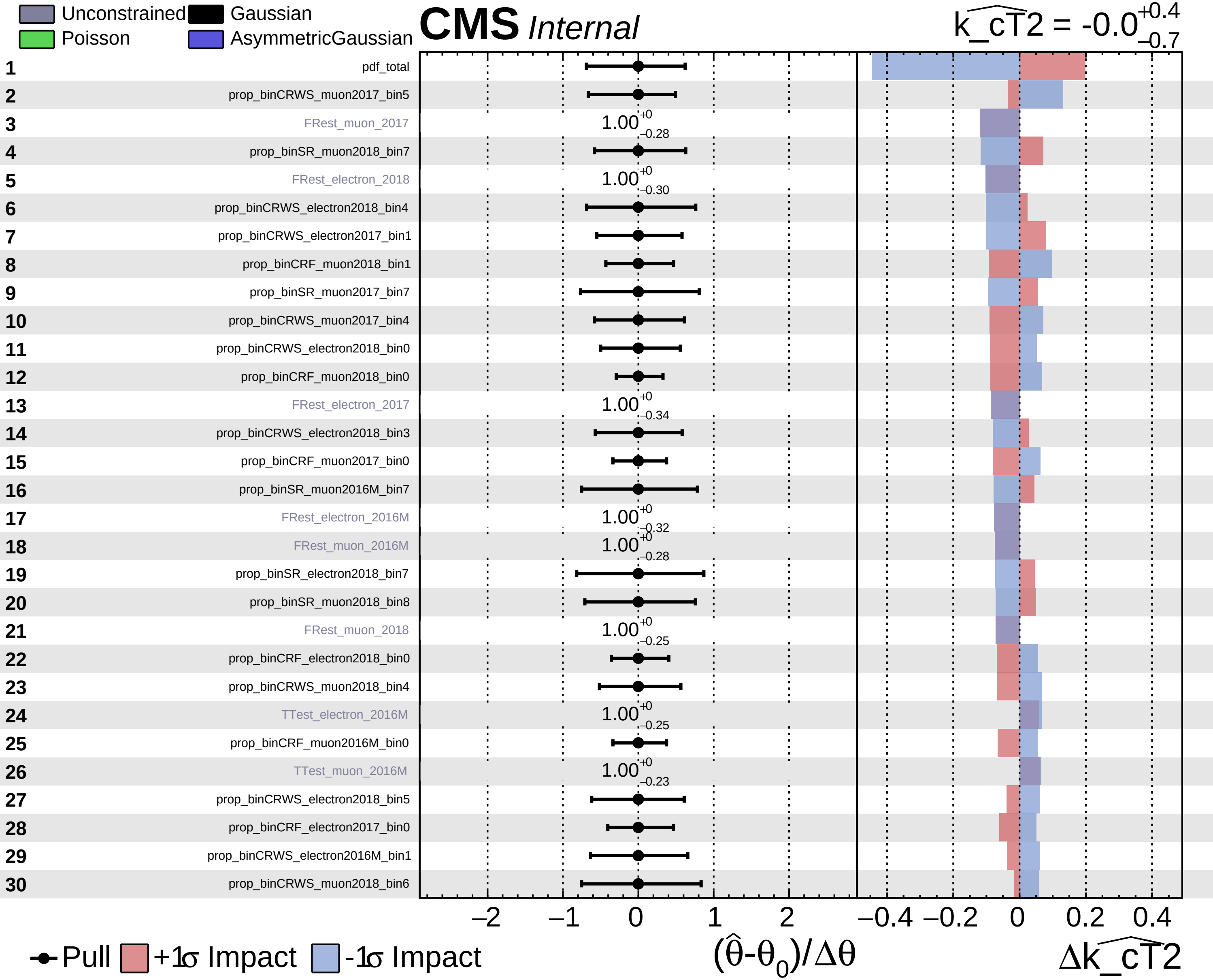
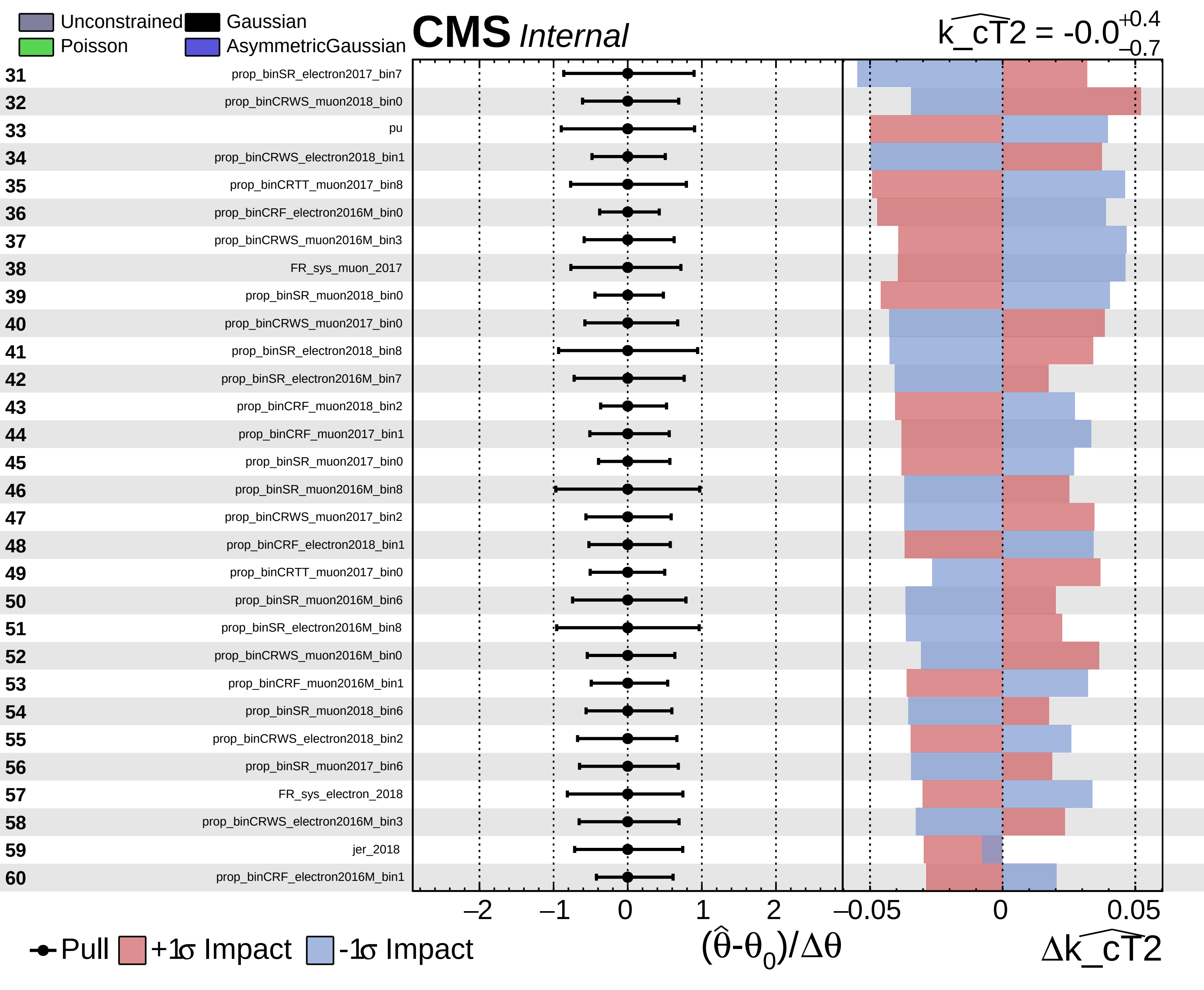
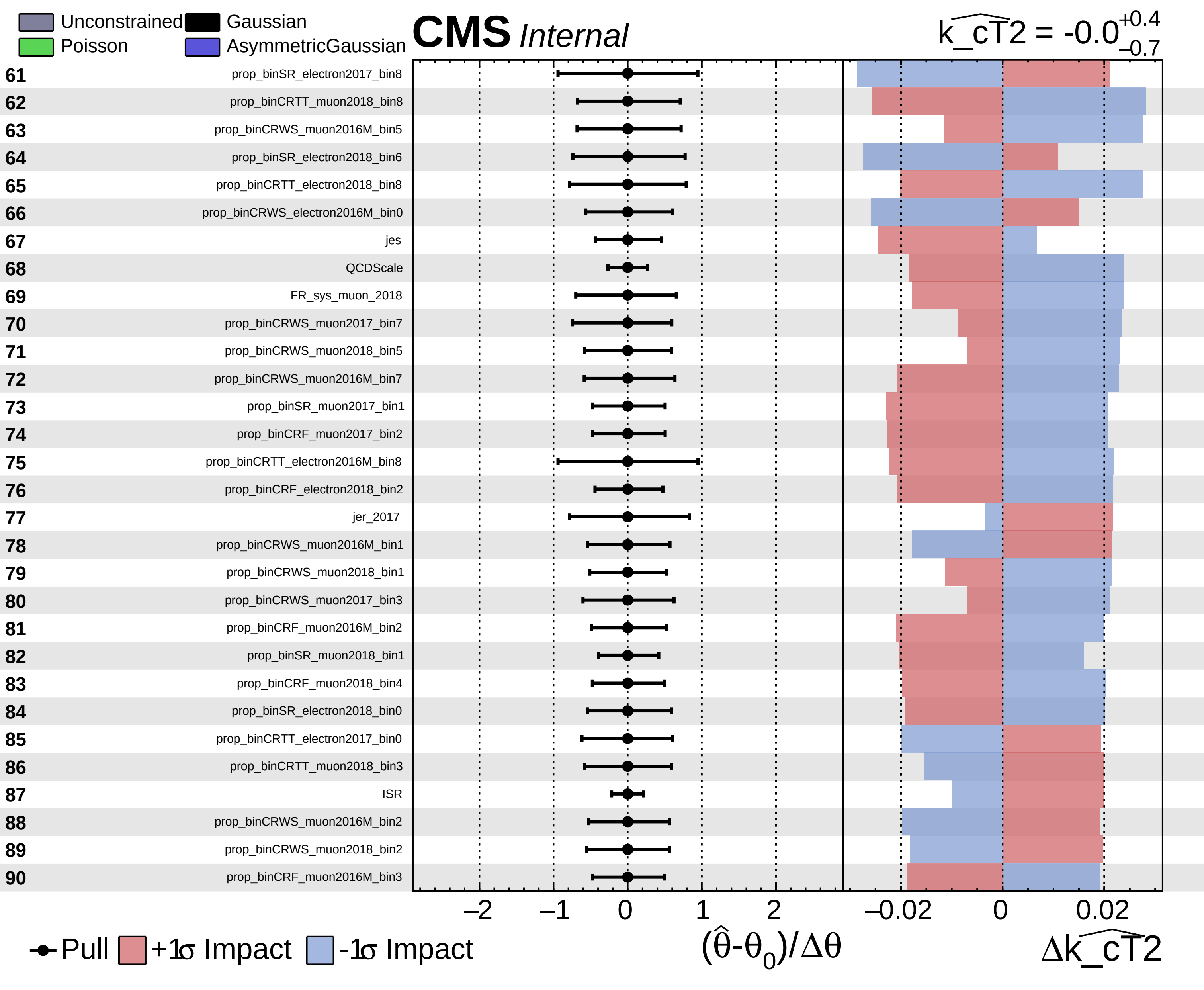


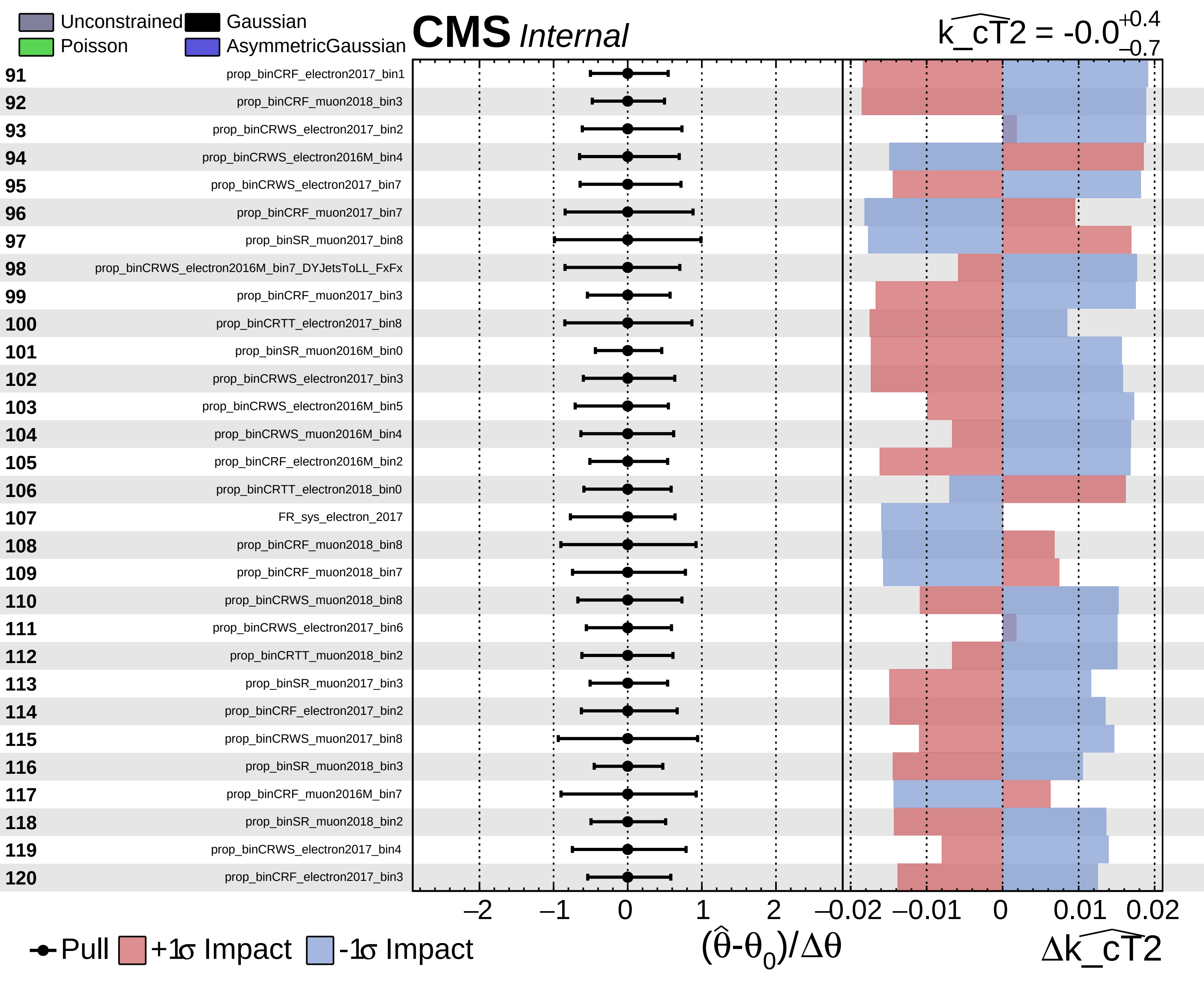
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.4 −0.7
1 pdf_total
2 prop_binCRWS_muon2017_bin5
3 FRest_muon_2017
1.00 +0 −0.28
4 prop_binSR_muon2018_bin7
5 FRest_electron_2018
1.00 +0 −0.30
6 prop_binCRWS_electron2018_bin4
7 prop_binCRWS_electron2017_bin1
8 prop_binCRF_muon2018_bin1
9 prop_binSR_muon2017_bin7
10 prop_binCRWS_muon2017_bin4
11 prop_binCRWS_electron2018_bin0
12 prop_binCRF_muon2018_bin0
13 FRest_electron_2017
1.00 +0 −0.34
14 prop_binCRWS_electron2018_bin3
15 prop_binCRF_muon2017_bin0
16 prop_binSR_muon2016M_bin7
17 FRest_electron_2016M
1.00 +0 −0.32
18 FRest_muon_2016M
1.00 +0 −0.28
19 prop_binSR_electron2018_bin7
20 prop_binSR_muon2018_bin8
21 FRest_muon_2018
1.00 +0 −0.25
22 prop_binCRF_electron2018_bin0
23 prop_binCRWS_muon2018_bin4
24 TTest_electron_2016M
1.00 +0 −0.25
25 prop_binCRF_muon2016M_bin0
26 TTest_muon_2016M
1.00 +0 −0.23
27 prop_binCRWS_electron2018_bin5
28 prop_binCRF_electron2017_bin0
29 prop_binCRWS_electron2016M_bin1
30 prop_binCRWS_muon2018_bin6
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT2

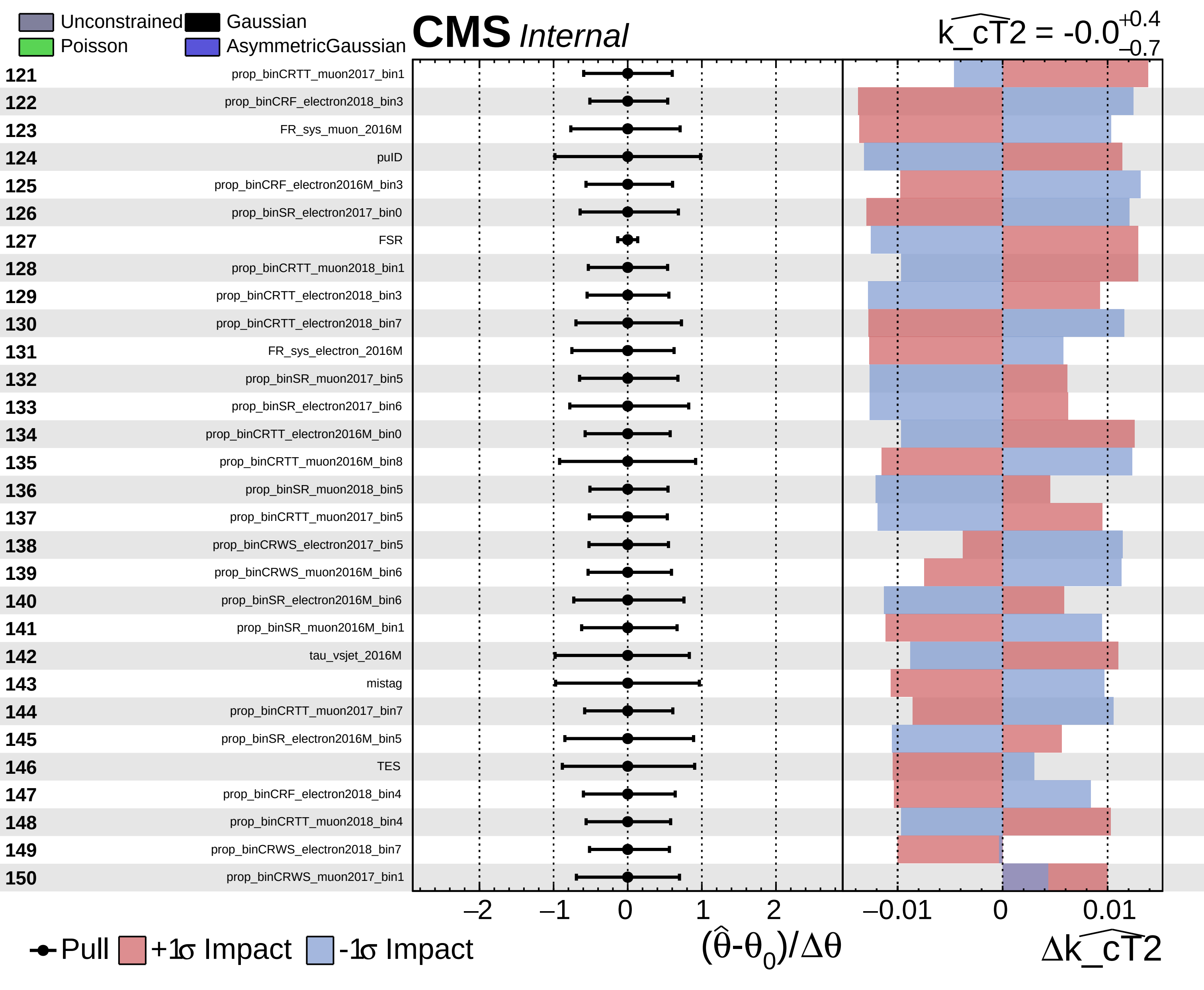


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.4 −0.7
31 prop_binSR_electron2017_bin7
32 prop_binCRWS_muon2018_bin0
33 pu
34 prop_binCRWS_electron2018_bin1
35 prop_binCRTT_muon2017_bin8
36 prop_binCRF_electron2016M_bin0
37 prop_binCRWS_muon2016M_bin3
38 FR_sys_muon_2017
39 prop_binSR_muon2018_bin0
40 prop_binCRWS_muon2017_bin0
41 prop_binSR_electron2018_bin8
42 prop_binSR_electron2016M_bin7
43 prop_binCRF_muon2018_bin2
44 prop_binCRF_muon2017_bin1
45 prop_binSR_muon2017_bin0
46 prop_binSR_muon2016M_bin8
47 prop_binCRWS_muon2017_bin2
48 prop_binCRF_electron2018_bin1
49 prop_binCRTT_muon2017_bin0
50 prop_binSR_muon2016M_bin6
51 prop_binSR_electron2016M_bin8
52 prop_binCRWS_muon2016M_bin0
53 prop_binCRF_muon2016M_bin1
54 prop_binSR_muon2018_bin6
55 prop_binCRWS_electron2018_bin2
56 prop_binSR_muon2017_bin6
57 FR_sys_electron_2018
58 prop_binCRWS_electron2016M_bin3
59 jer_2018
60 prop_binCRF_electron2016M_bin1
−2 −1 0 1 2
−0.05 0 0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT2

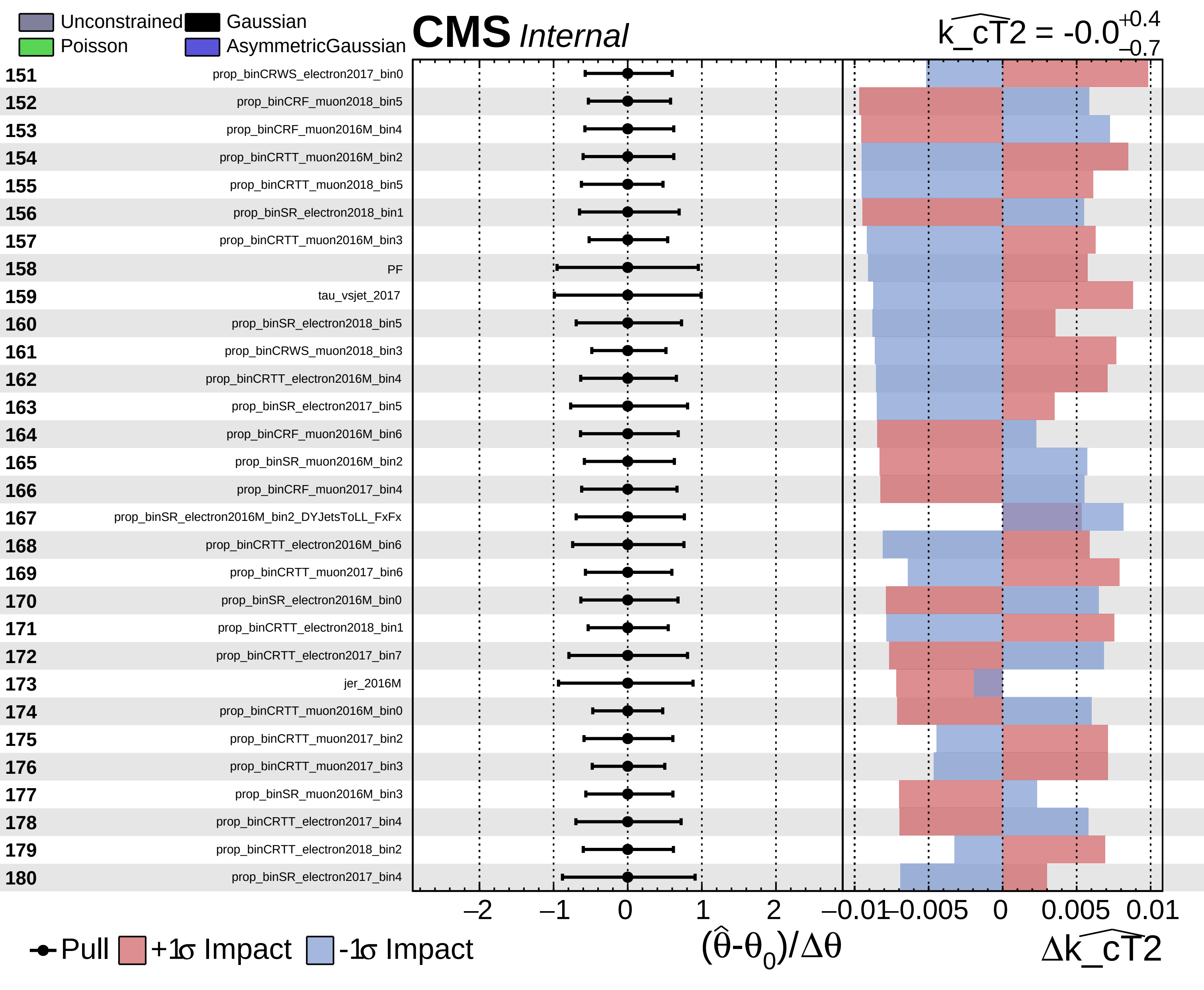


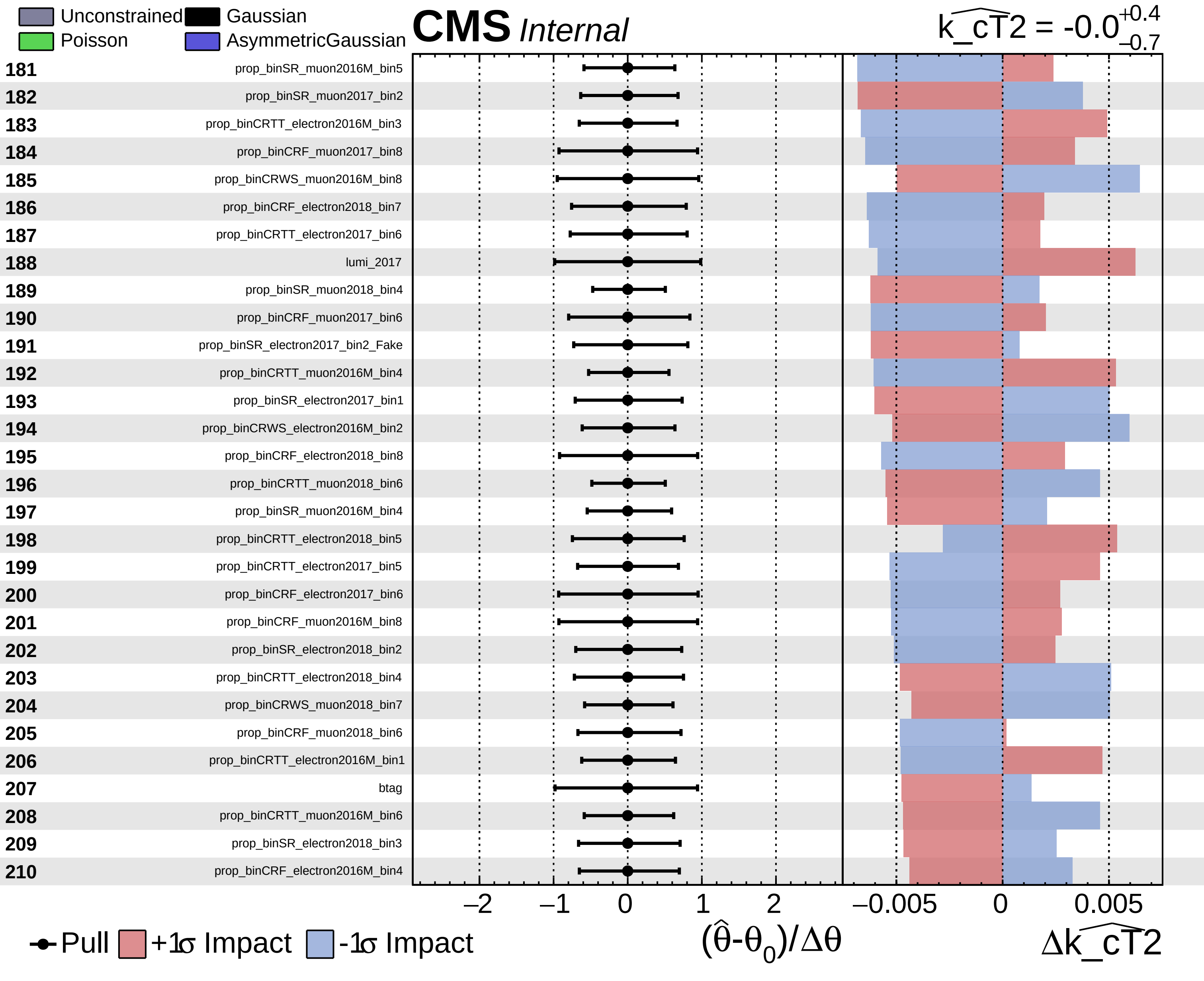
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.4 −0.7
61 prop_binSR_electron2017_bin8
62 prop_binCRTT_muon2018_bin8
63 prop_binCRWS_muon2016M_bin5
64 prop_binSR_electron2018_bin6
65 prop_binCRTT_electron2018_bin8
66 prop_binCRWS_electron2016M_bin0
67 jes
68 QCDScale
69 FR_sys_muon_2018
70 prop_binCRWS_muon2017_bin7
71 prop_binCRWS_muon2018_bin5
72 prop_binCRWS_muon2016M_bin7
73 prop_binSR_muon2017_bin1
74 prop_binCRF_muon2017_bin2
75 prop_binCRTT_electron2016M_bin8
76 prop_binCRF_electron2018_bin2
77 jer_2017
78 prop_binCRWS_muon2016M_bin1
79 prop_binCRWS_muon2018_bin1
80 prop_binCRWS_muon2017_bin3
81 prop_binCRF_muon2016M_bin2
82 prop_binSR_muon2018_bin1
83 prop_binCRF_muon2018_bin4
84 prop_binSR_electron2018_bin0
85 prop_binCRTT_electron2017_bin0
86 prop_binCRTT_muon2018_bin3
87 ISR
88 prop_binCRWS_muon2016M_bin2
89 prop_binCRWS_muon2018_bin2
90 prop_binCRF_muon2016M_bin3
−2 −1 0 1 2
−0.02 0 0.02
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = -0.0^{+0.4}_{-0.7}$
91 prop_binCRF_electron2017_bin1
92 prop_binCRF_muon2018_bin3
93 prop_binCRWS_electron2017_bin2
94 prop_binCRWS_electron2016M_bin4
95 prop_binCRWS_electron2017_bin7
96 prop_binCRF_muon2017_bin7
97 prop_binSR_muon2017_bin8
98 prop_binCRWS_electron2016M_bin7_DYJetsToLL_FxFx
99 prop_binCRF_muon2017_bin3
100 prop_binCRTT_electron2017_bin8
101 prop_binSR_muon2016M_bin0
102 prop_binCRWS_electron2017_bin3
103 prop_binCRWS_electron2016M_bin5
104 prop_binCRWS_muon2016M_bin4
105 prop_binCRF_electron2016M_bin2
106 prop_binCRTT_electron2018_bin0
107 FR_sys_electron_2017
108 prop_binCRF_muon2018_bin8
109 prop_binCRF_muon2018_bin7
110 prop_binCRWS_muon2018_bin8
111 prop_binCRWS_electron2017_bin6
112 prop_binCRTT_muon2018_bin2
113 prop_binSR_muon2017_bin3
114 prop_binCRF_electron2017_bin2
115 prop_binCRWS_muon2017_bin8
116 prop_binSR_muon2018_bin3
117 prop_binCRF_muon2016M_bin7
118 prop_binSR_muon2018_bin2
119 prop_binCRWS_electron2017_bin4
120 prop_binCRF_electron2017_bin3
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$

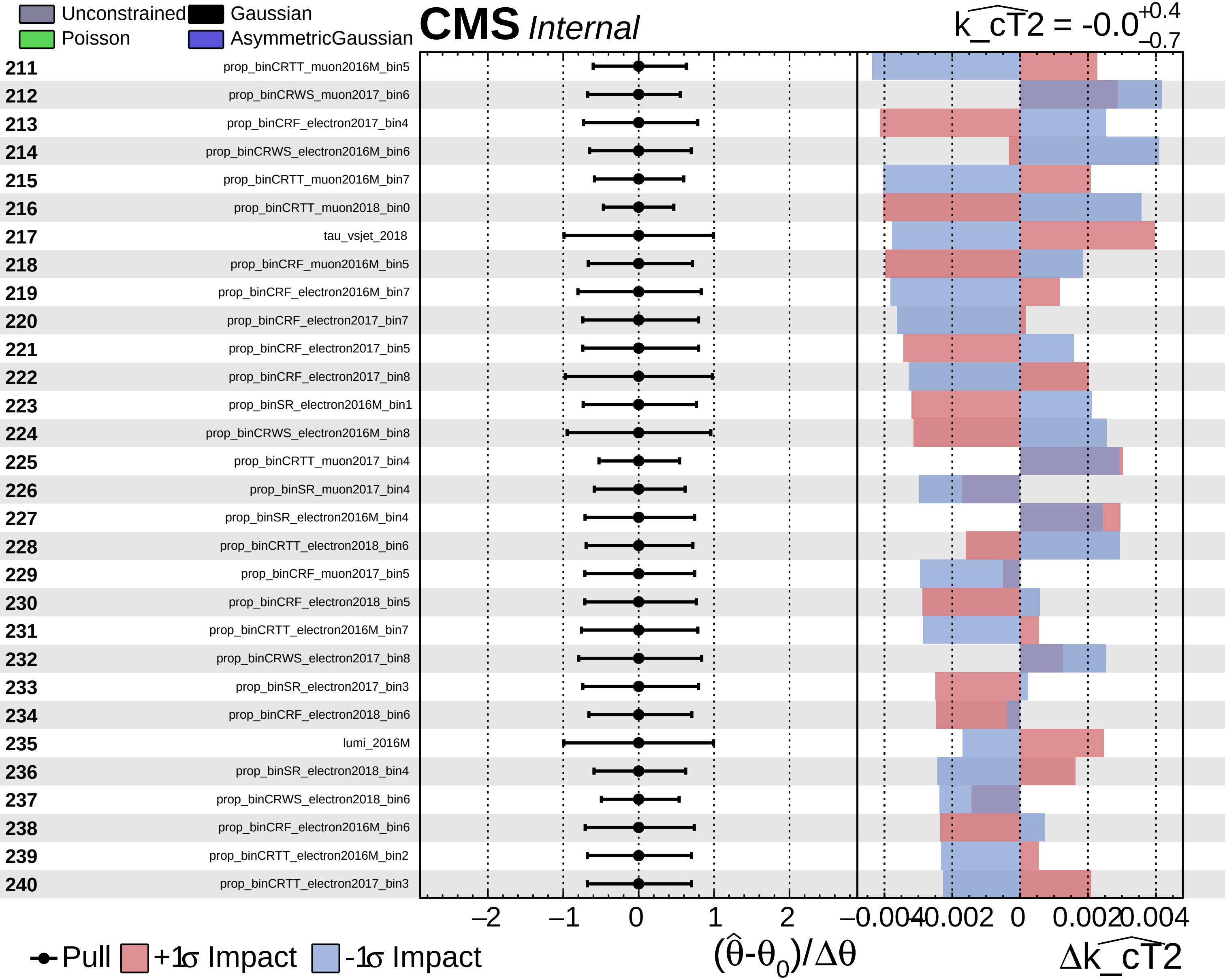


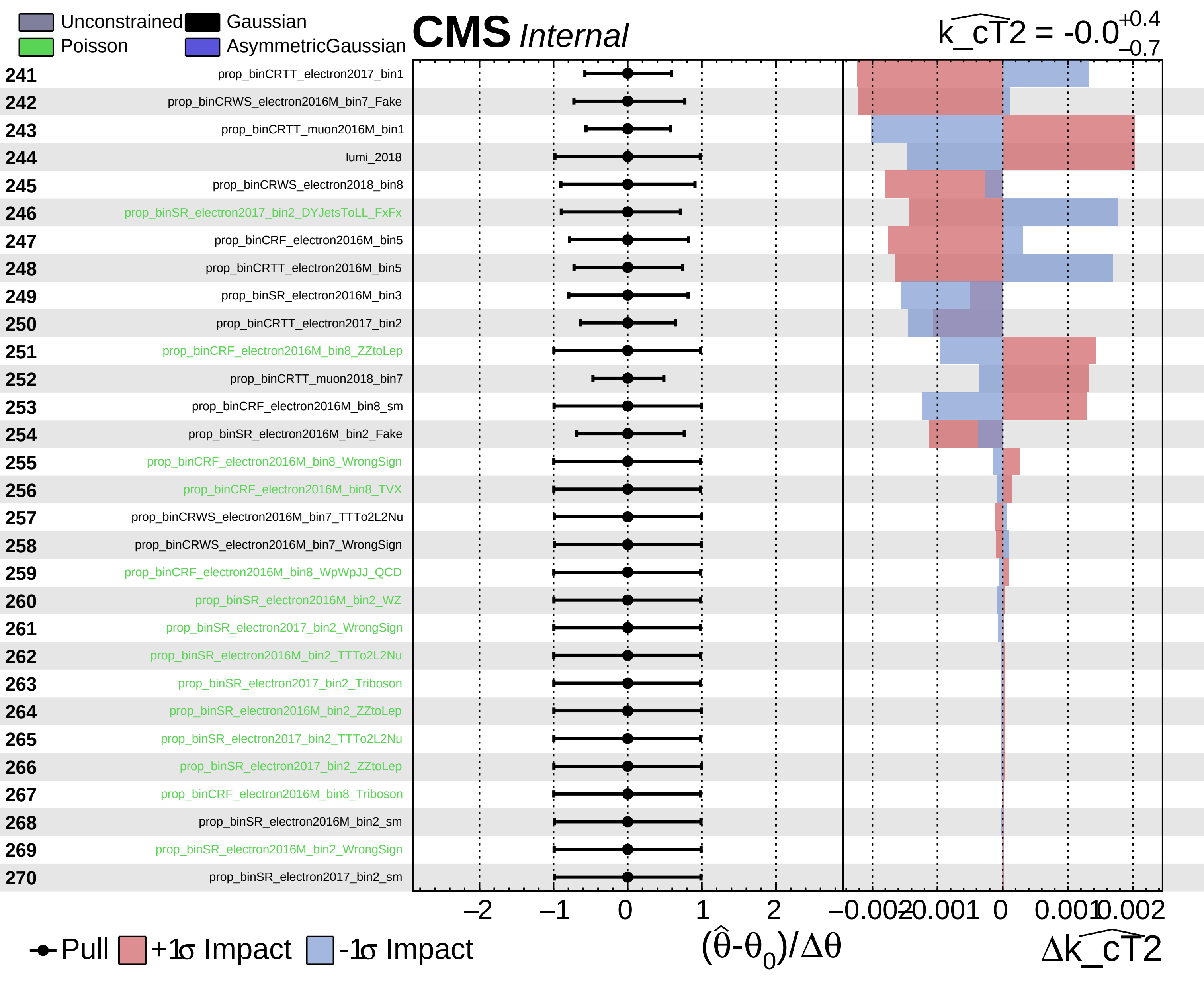
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.4 −0.7
121 prop_binCRTT_muon2017_bin1
122 prop_binCRF_electron2018_bin3
123 FR_sys_muon_2016M
124 puID
125 prop_binCRF_electron2016M_bin3
126 prop_binSR_electron2017_bin0
127 FSR
128 prop_binCRTT_muon2018_bin1
129 prop_binCRTT_electron2018_bin3
130 prop_binCRTT_electron2018_bin7
131 FR_sys_electron_2016M
132 prop_binSR_muon2017_bin5
133 prop_binSR_electron2017_bin6
134 prop_binCRTT_electron2016M_bin0
135 prop_binCRTT_muon2016M_bin8
136 prop_binSR_muon2018_bin5
137 prop_binCRTT_muon2017_bin5
138 prop_binCRWS_electron2017_bin5
139 prop_binCRWS_muon2016M_bin6
140 prop_binSR_electron2016M_bin6
141 prop_binSR_muon2016M_bin1
142 tau_vsjet_2016M
143 mistag
144 prop_binCRTT_muon2017_bin7
145 prop_binSR_electron2016M_bin5
146 TES
147 prop_binCRF_electron2018_bin4
148 prop_binCRTT_muon2018_bin4
149 prop_binCRWS_electron2018_bin7
150 prop_binCRWS_muon2017_bin1
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.4 −0.7
151 prop_binCRWS_electron2017_bin0
152 prop_binCRF_muon2018_bin5
153 prop_binCRF_muon2016M_bin4
154 prop_binCRTT_muon2016M_bin2
155 prop_binCRTT_muon2018_bin5
156 prop_binSR_electron2018_bin1
157 prop_binCRTT_muon2016M_bin3
158 PF
159 tau_vsjet_2017
160 prop_binSR_electron2018_bin5
161 prop_binCRWS_muon2018_bin3
162 prop_binCRTT_electron2016M_bin4
163 prop_binSR_electron2017_bin5
164 prop_binCRF_muon2016M_bin6
165 prop_binSR_muon2016M_bin2
166 prop_binCRF_muon2017_bin4
167 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
168 prop_binCRTT_electron2016M_bin6
169 prop_binCRTT_muon2017_bin6
170 prop_binSR_electron2016M_bin0
171 prop_binCRTT_electron2018_bin1
172 prop_binCRTT_electron2017_bin7
173 jer_2016M
174 prop_binCRTT_muon2016M_bin0
175 prop_binCRTT_muon2017_bin2
176 prop_binCRTT_muon2017_bin3
177 prop_binSR_muon2016M_bin3
178 prop_binCRTT_electron2017_bin4
179 prop_binCRTT_electron2018_bin2
180 prop_binSR_electron2017_bin4
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

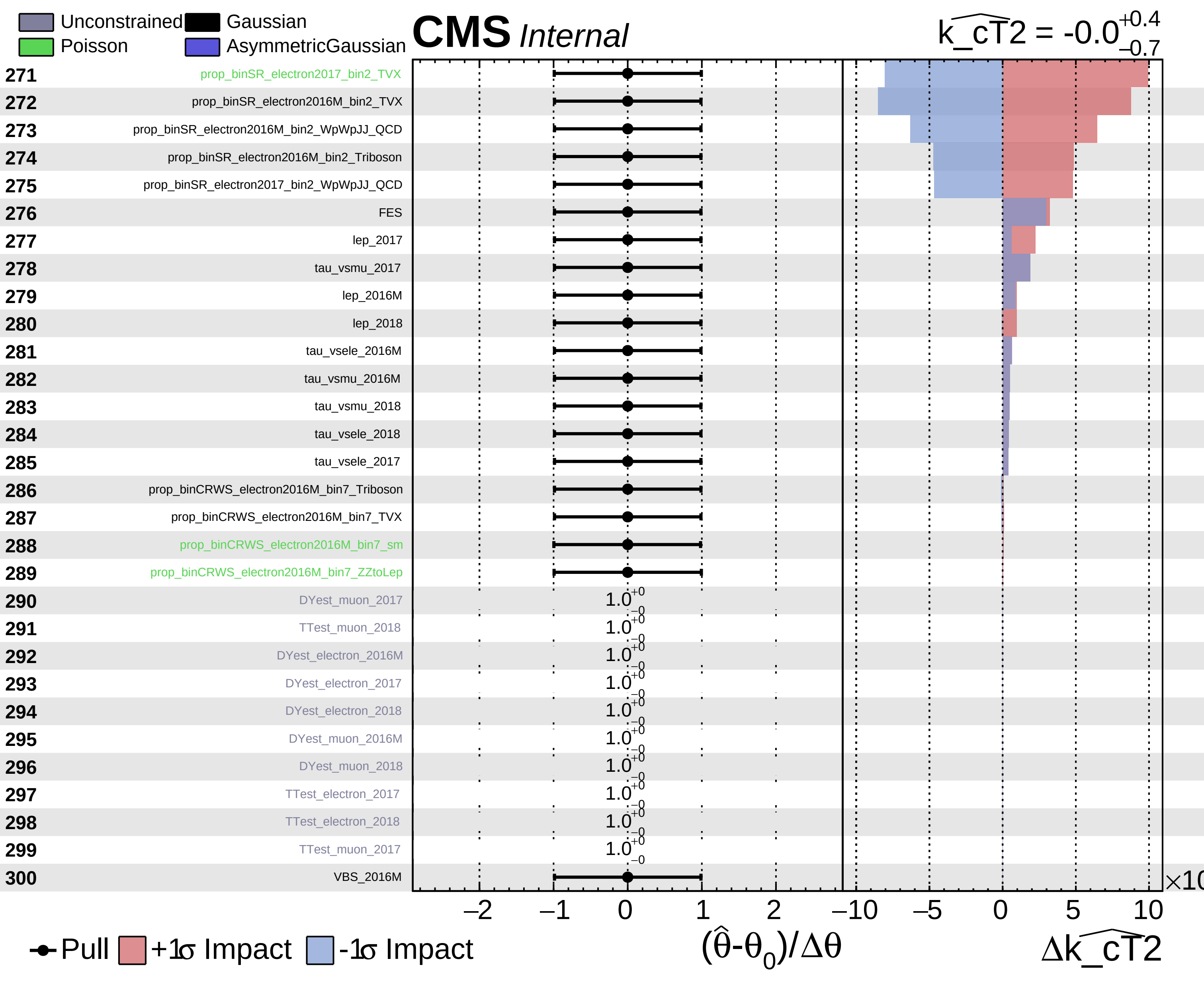


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = -0.0^{+0.4}_{-0.7}$
181 prop_binSR_muon2016M_bin5
182 prop_binSR_muon2017_bin2
183 prop_binCRTT_electron2016M_bin3
184 prop_binCRF_muon2017_bin8
185 prop_binCRWS_muon2016M_bin8
186 prop_binCRF_electron2018_bin7
187 prop_binCRTT_electron2017_bin6
188 lumi_2017
189 prop_binSR_muon2018_bin4
190 prop_binCRF_muon2017_bin6
191 prop_binSR_electron2017_bin2_Fake
192 prop_binCRTT_muon2016M_bin4
193 prop_binSR_electron2017_bin1
194 prop_binCRWS_electron2016M_bin2
195 prop_binCRF_electron2018_bin8
196 prop_binCRTT_muon2018_bin6
197 prop_binSR_muon2016M_bin4
198 prop_binCRTT_electron2018_bin5
199 prop_binCRTT_electron2017_bin5
200 prop_binCRF_electron2017_bin6
201 prop_binCRF_muon2016M_bin8
202 prop_binSR_electron2018_bin2
203 prop_binCRTT_electron2018_bin4
204 prop_binCRWS_muon2018_bin7
205 prop_binCRF_muon2018_bin6
206 prop_binCRTT_electron2016M_bin1
207 btag
208 prop_binCRTT_muon2016M_bin6
209 prop_binSR_electron2018_bin3
210 prop_binCRF_electron2016M_bin4
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$

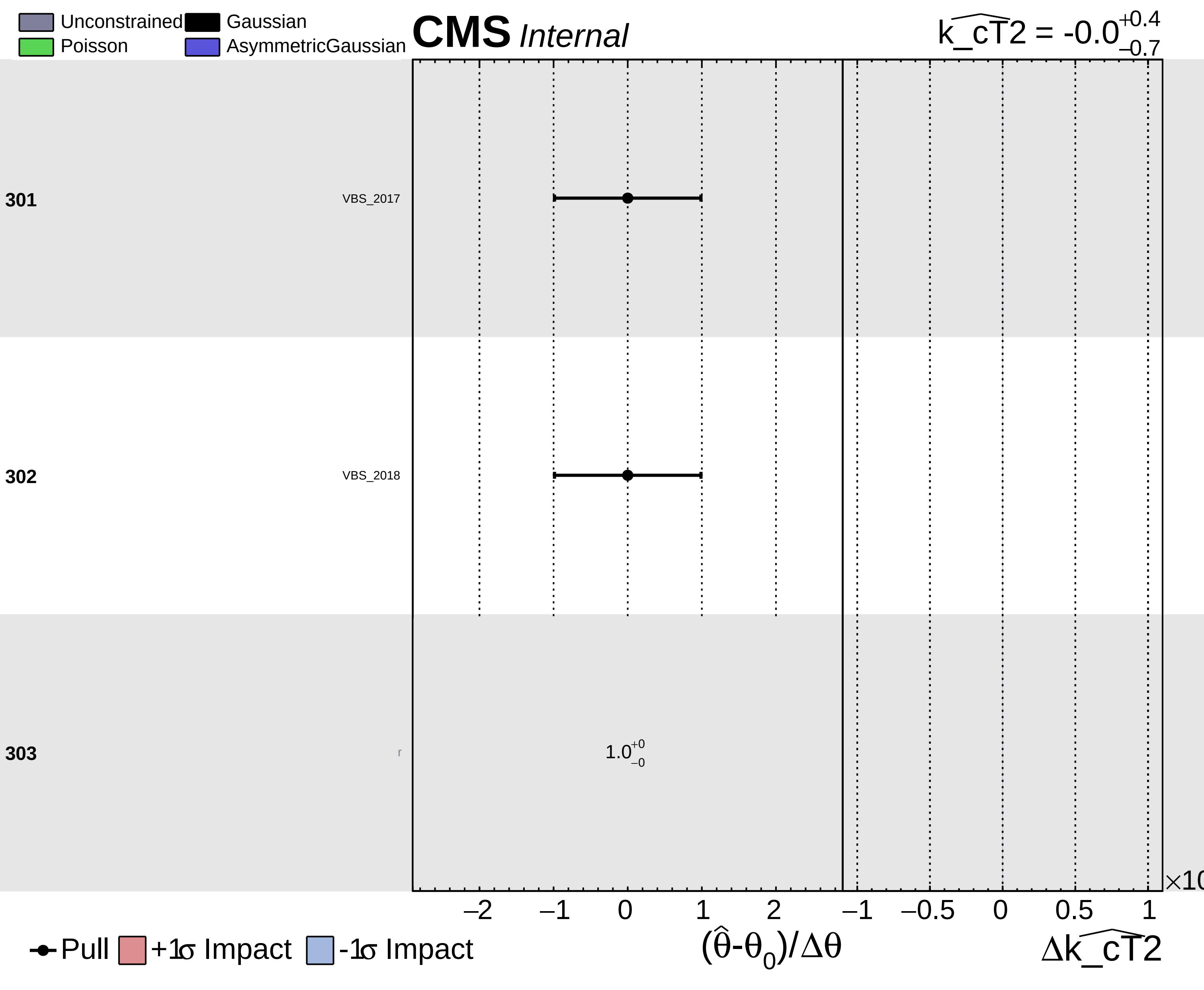

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.4 −0.7
211 prop_binCRTT_muon2016M_bin5
212 prop_binCRWS_muon2017_bin6
213 prop_binCRF_electron2017_bin4
214 prop_binCRWS_electron2016M_bin6
215 prop_binCRTT_muon2016M_bin7
216 prop_binCRTT_muon2018_bin0
217 tau_vsjet_2018
218 prop_binCRF_muon2016M_bin5
219 prop_binCRF_electron2016M_bin7
220 prop_binCRF_electron2017_bin7
221 prop_binCRF_electron2017_bin5
222 prop_binCRF_electron2017_bin8
223 prop_binSR_electron2016M_bin1
224 prop_binCRWS_electron2016M_bin8
225 prop_binCRTT_muon2017_bin4
226 prop_binSR_muon2017_bin4
227 prop_binSR_electron2016M_bin4
228 prop_binCRTT_electron2018_bin6
229 prop_binCRF_muon2017_bin5
230 prop_binCRF_electron2018_bin5
231 prop_binCRTT_electron2016M_bin7
232 prop_binCRWS_electron2017_bin8
233 prop_binSR_electron2017_bin3
234 prop_binCRF_electron2018_bin6
235 lumi_2016M
236 prop_binSR_electron2018_bin4
237 prop_binCRWS_electron2018_bin6
238 prop_binCRF_electron2016M_bin6
239 prop_binCRTT_electron2016M_bin2
240 prop_binCRTT_electron2017_bin3
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.4 −0.7
241 prop_binCRTT_electron2017_bin1
242 prop_binCRWS_electron2016M_bin7_Fake
243 prop_binCRTT_muon2016M_bin1
244 lumi_2018
245 prop_binCRWS_electron2018_bin8
246 prop_binSR_electron2017_bin2_DYJetsToLL_FxFx
247 prop_binCRF_electron2016M_bin5
248 prop_binCRTT_electron2016M_bin5
249 prop_binSR_electron2016M_bin3
250 prop_binCRTT_electron2017_bin2
251 prop_binCRF_electron2016M_bin8_ZZtoLep
252 prop_binCRTT_muon2018_bin7
253 prop_binCRF_electron2016M_bin8_sm
254 prop_binSR_electron2016M_bin2_Fake
255 prop_binCRF_electron2016M_bin8_WrongSign
256 prop_binCRF_electron2016M_bin8_TVX
257 prop_binCRWS_electron2016M_bin7_TTTo2L2Nu
258 prop_binCRWS_electron2016M_bin7_WrongSign
259 prop_binCRF_electron2016M_bin8_WpWpJJ_QCD
260 prop_binSR_electron2016M_bin2_WZ
261 prop_binSR_electron2017_bin2_WrongSign
262 prop_binSR_electron2016M_bin2_TTTo2L2Nu
263 prop_binSR_electron2017_bin2_Triboson
264 prop_binSR_electron2016M_bin2_ZZtoLep
265 prop_binSR_electron2017_bin2_TTTo2L2Nu
266 prop_binSR_electron2017_bin2_ZZtoLep
267 prop_binCRF_electron2016M_bin8_Triboson
268 prop_binSR_electron2016M_bin2_sm
269 prop_binSR_electron2016M_bin2_WrongSign
270 prop_binSR_electron2017_bin2_sm
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.4 −0.7
271 prop_binSR_electron2017_bin2_TVX
272 prop_binSR_electron2016M_bin2_TVX
273 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
274 prop_binSR_electron2016M_bin2_Triboson
275 prop_binSR_electron2017_bin2_WpWpJJ_QCD
276 FES
277 lep_2017
278 tau_vsmu_2017
279 lep_2016M
280 lep_2018
281 tau_vsele_2016M
282 tau_vsmu_2016M
283 tau_vsmu_2018
284 tau_vsele_2018
285 tau_vsele_2017
286 prop_binCRWS_electron2016M_bin7_Triboson
287 prop_binCRWS_electron2016M_bin7_TVX
288 prop_binCRWS_electron2016M_bin7_sm
289 prop_binCRWS_electron2016M_bin7_ZZtoLep
290 DYest_muon_2017 1.0 +0 −0
291 TTest_muon_2018 1.0 +0 −0
292 DYest_electron_2016M 1.0 +0 −0
293 DYest_electron_2017 1.0 +0 −0
294 DYest_electron_2018 1.0 +0 −0
295 DYest_muon_2016M 1.0 +0 −0
296 DYest_muon_2018 1.0 +0 −0
297 TTest_electron_2017 1.0 +0 −0
298 TTest_electron_2018 1.0 +0 −0
299 TTest_muon_2017 1.0 +0 −0
300 VBS_2016M
−2 −1 0 1 2
−10 −5 0 5 10
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.4 −0.7
301
VBS_2017
302
VBS_2018
303
r
1.0 +0 −0
×10
−2 −1 0 1 2
−1 −0.5 0 0.5 1
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2